

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Computer Applications (MCA)

Semester – I University Examination 2009 – 10

CA702 Fundamentals of Computer System Organization

Date: 26/12/09, Saturday

Time: 1:00 p.m to 4:00 p.m

Maximum Marks: 70

SECTION-I

Q1 Do as directed:

(A) Answer the following questions. (Any 5.)

[10]

(1) What are the values of x, y, and z.

$$(1011)_2 = (x)_{10} = (y)_8 = (z)_{16}$$

(2) Perform decimal subtraction, using 9s & 10s complement:
8-8

(3) $(25)_8 + (63)_8$ in Octal system

(4) $(5)_7 * (25)_7$ in base-7 system

(5) Represent decimal number 123 in BCD code and Excess-3 code

(6) Convert 010110 into gray code and gray code to again binary

(7) Formulate a weighted binary code for the decimal digits using XS-3.

(8) What are the various ways to represent Negative Numbers in the Computer system?

(B) Write a short note on Excess – 3 code with proper example.

[2]

(C) Answer the following questions:

[3]

(1) Which gates are called Universal Gates? Why?

(2) Realize X-NOR gate using NAND gate.

Q2 Do as directed:

(A) Explain the JK flip-flop with circuit diagram, characteristic table and characteristic equation.

[5]

(B) Design an AND-to-OR gate combinational network for the Boolean algebra expression

[3]

$$AB'CD + ABC'D + A'B'CD + ABC'D' + AB'C'D' + A'BCD$$

(C) Draw the timing diagram for each output of the 3 bit shift register, if the input sequence is 101100 in accordance with 5 clock pulses.

[2]

OR

Q2 Do as directed:

(A) Explain Master-Slave Flip-flop with block diagram and logic design.

[5]

(B) Explain the transfer circuit with the circuit diagram.

[3]

(C) What do you mean by "Clock Pulse"? Explain in brief.

[2]

Q3 Do as directed:

(A) Prove the following using Boolean Algebra:

[4]

$$(1) A'C + A'B + AB'C + BC = C + A'B$$

$$(2) (A+B+C)(A+B'+C')(A+B+C')(A+B'+C)=A$$

(B) Use as few gates as possible, design a NAND-to-AND gate network that realizes the following Boolean algebra expression.

[3]

$$A'BC'D + ABC'D' + A'B'CD' + AB'C'D'$$

(C) What is a flip-flop? Explain RS flip-flop.

[3]

OR

- Q3 Do as directed:**
- (A) $F(W, X, Y, Z) = \sum (4, 6, 7, 8) + D(2, 5, 11, 12)$ using K-Map [6]
1. Find SOP expression
 2. Simplify SOP expression
 3. Implement this simplified expression using two level AND-to-OR gate network
 4. Implement this simplified expression using NAND gates only
- (B) Draw Karnaugh maps in W, X, Y, Z. Also, find the subcubes. $M1+M3+M5+M7+M12+M13+M8+M9$ [3]
- (C) Perform the binary multiplication on following: $15 * 2$ [1]

Section II

- Q4 Do as directed:**
- (A) Explain Half Adder. [4]
- (B) Write a short note on DMA. [5]
- (C) Explain NAND Gate Decoder with circuit diagram. [4]
- (D) Find the complement of $B'D + A'BC' + ACD + A'BC$, also simplify it. [2]
- Q5 Do as directed:**
- (A) Perform multiplication with showing the contents of accumulator, B register and Y register during each step. $B=04$ $Y=03$ [5]
- (B) Which operations are needed to perform PUSH or POP operation in stack? [2]
- (C) Write a short note on BCD adder. [3]

OR

- Q5 Do as directed:**
- (A) Perform division with showing the contents of accumulator, B register and Y register during each step. $B=7$ $Y=3$ [5]
- (B) Which are the different registers used in register stack? Explain the functionality of it. [2]
- (C) What is an Instruction Format? What is a field? Which are the fields in Instruction Format? [3]
- Q6 Do as directed:**
- (A) Evaluate the following arithmetic operation using RPN. $(16*2) / (12/3) - (2*3)$. Also list out steps for it. [4]
- (B) Explain the characteristics of printer. Differentiate between Impact and non Impact printer. [4]
- (C) What is an Asynchronous Serial Transfer? Explain in brief. [2]

OR

- Q6 Do as directed:**
- (A) What is an Asynchronous Data Transfer? Explain Handshaking method of it. [4]
- (B) Explain in brief: Cache memory, Virtual memory [4]
- (C) (1) Which are the different types of ROM? [2]
(2) Which are the operating modes available for RAM chips? What do they consist?